TED (21) – REVISION 2021

FOURTH SEMESTER DIPLOMA EXAMINATION IN
ENGINEERING AND TECHNOLOGY

THERAPEUTIC EQUIPMENT

(MODEL QUESTION PAPER)

Time: 3 hours Maximum Marks: 75

PART A

I. Answer all questions in one word or one sentence. Each question carries one mark.

(9 x 1 = 9 Marks)

1	Activity of which chamber is sensed in ventricular synchronous demand pacemaker?	M1.01	U
2	Name the type of waveform used in DC defibrillator?	M1.02	R
3	Draw the blend waveform used for electrosurgery?	M2.01	U
4	What is the principle of ultrasonic lithotripsy?	M2.02	R
5	Ventilator is operated in which mode for a patient who is able to initiate an inspiration?	M3.01	U
6	Give an example for a condition in which cardiopulmonary bypass is used?	M3.03	R
7	Mention two important parameters that can be adjusted in an infusion pump?	M4.01	R
8	Name any two application specific endoscopes with their application?	M4.02	R
9	The principle of light conduction in endoscope is ………………….?	M4.03	R

PART B

II. Answer any eight questions from the following. Each question carries 3 marks

(8 x 3 = 24 Marks)

1	What are the techniques used for programming an implantable pacemaker?	M1.01	U
2	Differentiate between the emergency mode and synchronized mode of defibrillator?	M1.02	U

3	With diagram explain the process of cutting and coagulation?	M2.01	U
4	Draw the diagram of a hollow fibre type haemodialyzer?	M2.03	R
5	What is cycling in ventilators and outline the different types of cycling?	M3.01	U
6	What is the role of vaporizers in anesthesia machine, also give the mechanisms for anesthetic gas flow?	M3.02	U
7	What is an isodose chart and give its significance in radio therapy?	M4.04	U
8	What are drug infusion sytems, explain their applications?	M4.01	R
9	What changes occur in human body when exposed to radiations while doing therapy?	M4.04	U
10	What are the different types of peristaltic drug infusion systems?	M4.01	R

PART C

Answer all questions. Each question carries seven marks

(6 x 7 = 42 Marks)

III	Draw the block diagram and explain the working of an implantable pacemaker?	**M1.01**	R
	OR		
IV	Describe the working of a cardioverter?	**M1.02**	U
V	Summarize the working of an ESU with schematic?	**M2.01**	U
	OR		
VI	Illustrate the system arrangement for ESWL and explain its working?	**M2.02**	U

VII	Explain the working of a typicalventilatorwith block diagram?	**M3.01**	U
	OR		
VIII	Describe the working of an anesthesiamachine?	**M3.02**	U

IX X	Illustrate the parts of an endoscope? OR Explain the principle and working of LINAC?	**M4.03** **M4.04**	R U
XI XII	Identify the most suitable power source for an implantable pacemaker justify your answer with a comparison of other power sources. OR How pain relief can be achieved through electrical stimulation, explain?	**M1.01** **M1.03**	U U
XIII XIV	Describe the working of a heartlung machine with block diagram? OR Explain different modes of operation of a ventilator?	**M3.03** **M3.01**	U R